# Sessions and Cookies management in PHP

# Topics

- What are cookies?
- Create and use cookies.
- What are sessions?
- Create and use sessions

# Introduction

- HTTP is a stateless protocol.
- A stateless protocol does not require the server to retain information or status about each user for the duration of multiple requests.
- this means that after an exchange is over...
  - a browser requests a resource from a server
  - the web server sends the resource to the browser
- ...the connection will be closed and forgotten
- this has its advantages
  - because there is nothing to be kept track of, it is relatively easy to build web servers that are very efficient
- But also it has drawbacks
  - it makes it hard to follow a user on a website

# Why to follow a user's?

- It is often extremely useful to be able to follow a user's activities on a website so that one can for example
  - have a shopping cart
  - maintain the user's identity
  - display information specifically tailored to the individual user
- Increased ability to provide the user a richer experience of using that website
  - a fine example of a site that truly exploits the user's identity is Amazon, which in many ways has set the standard in the application of user identity to provide a rich experience

# Login…

- Another aspect is the ability to log into a site with username and password
- this provides some level of security
- One gets the opportunity to have "your own" page
  - with personal things, like your own photos on Facebook
  - a personal configuration of the page
  - a user identity for postings on web boards
- etc.

- Some of the scenarios do not need to know "who" you are
  - "a specific browser on a particular machine" is often enough
  - here we often talk about the temporary nature of information
- In other situations it is useful to know "who" people are
  - one can get a more personalized experience of the site (this helps perhaps also with loyalty)
  - one can log on from different machines and have the same user experience

# So what can we do?

- One can add a parameter to the URL and remember it so that all subsequent links on the site contain it
  - but that gives ugly URLs that are difficult to maintain (one always has to rewrite all the URLs in a document) and are vulnerable to trivial hacks
- Since the web server can see, where a request comes from, one could use the user's machine address as an ID
  - but what if it is a shared computer?
  - or if it just looks like one computer due to NAT or a proxy?
- It would be great if a website could save a little bit of data on the user's machine ...

# So…

- So, web applications need to track the user's progress from page to page, for example when a web server is required to customize the content of a web page for a user.
- Solutions to address the above mentioned problems are cases that include:
  - the use of HTTP cookies.
  - server side sessions,
  - hidden variables (when the current page contains a form), and
  - URL-rewriting using URI-encoded parameters, e.g., /index.php?session_id=some_unique_session_code.

# Solution…

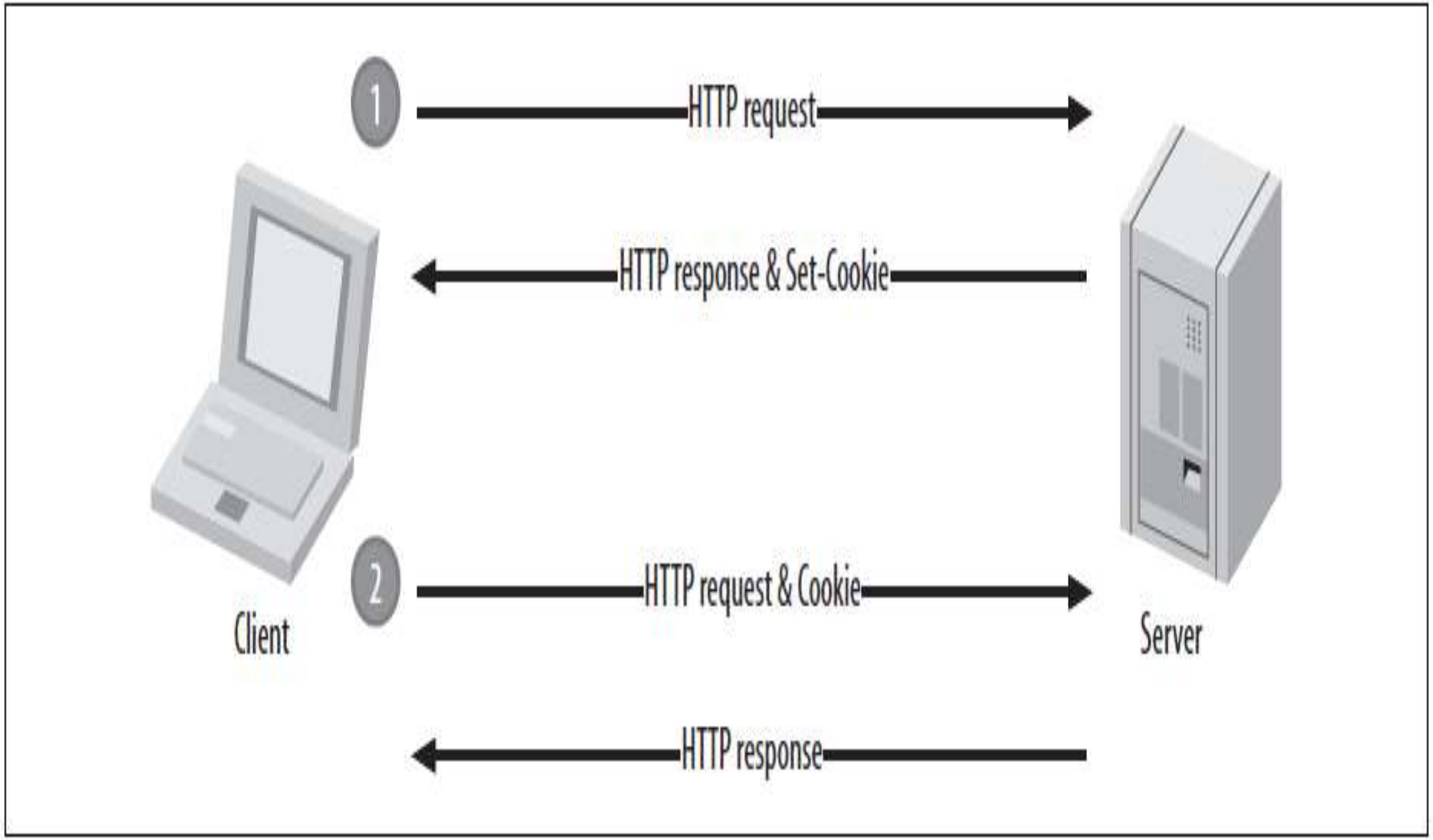
- HTTP remains stateless – there is no fixed connection between web server and browser
- While the stateless nature of HTTP has some important benefits
  - after all, maintaining state requires some overhead
  - it presents a unique challenge to developers who need to create stateful web applications.
- With no way to identify the client, it is impossible to determine
  - whether the user is already logged in,
  - has items in a shopping cart, or needs to register.
- An elegant solution to this problem, originally conceived by Netscape, is a state management mechanism called cookies.

# Cookies

- a web server can leave a "cookie" in the browser (i.e. on the user's computer)
  - it is up to the browser to manage these cookies
  - the cookie gets transmitted to the server in future connections
- A cookie is a small piece of data (typically max. 4 kB, usually far less) that is used by the web server to identify the user
- Cookies may be limited in time with an expiration date
  - else the cookie will be deleted when the browser is closed

# Cookies

- are an extension of the HTTP protocol.
- they consist of two HTTP headers:
  - the Set-Cookie response header and
  - the Cookie request header.
- When a client sends a request for a particular URL, the server can opt to include a Set-Cookie header in the response, so as to request for the client to include a corresponding Cookie header in its future requests
- Cookies
  - allow a unique identifier to be included in each request (in a Cookie header),
  - This help to uniquely identify clients and associate their requests together.




*A complete cookie exchange that involves two HTTP transactions*

# Create Cookies in PHP

- Use setcookie() function to create a cookie.

**Syntax**

**setcookie**(string *name, string value, int expire, string path, string domain, int secure*);

Example:

```
<?php
// Setting a cookie
setcookie("username",“Abebe", time()+10*24*60*60);
?>
```

| Parameter | Description |
| --- | --- |
| name | The name of the cookie. |
| value | The value of the cookie. |
| expires | The expiry date in UNIX timestamp format.<br>This implies, After this time cookie will become inaccessible.<br>The default value is 0. |
| path | Specify the path on the server for which the cookie will be available.<br>If set to '/', the cookie will be available within the entire domain. |
| domain | Specify the domain for which the cookie is available to e.g www.example.com. |
| secure | This field, if present, indicates that the cookie should be sent only if a secure HTTPS connection exists.<br>Is a Keyword referring that the cookie will not be sent over a plain HTTP connection |

```php
if(!empty($_POST["remember"]))
{
    setcookie ("member_login",$name,time()+ (10 * 365 * 24 * 60 * 60));
    setcookie ("member_password",$Password,time()+ (10 * 365 * 24 * 60 * 60));
}
else
{
    if(isset($_COOKIE["member_login"]))
    {
        setcookie ("member_login","");
    }
    if(isset($_COOKIE["member_password"]))
    {
        setcookie ("member_password","");
    }
}
```

# Accessing cookie

- The PHP $_COOKIE super global variable is used to retrieve a cookie value.

```
<html>
<head><title>sample on cookie</title></head>
<body>
<?php
if(!isset($_COOKIE["username"]))
{

setcookie("username", "Abebe", time()+10*24*60*60);
}
else{
echo $_COOKIE["username"]; // used to access a cookie
}?>
</body>
</html>
```

# Check if Cookies are Enabled

- count the $_COOKIE array variable

```
<?php
setcookie("username", "Abebe", time()+10*24*60*60);
?>
<html>
<body>

<?php
if(count($_COOKIE) > 0) {
    echo "Cookies are enabled.";
} else {
    echo "Cookies are disabled.";
}
?>
</body>
</html>
```

# Removing Cookies

Cookies can be deleted by calling the ***setcookie()*** function with the cookie name and any value (such as an empty string) with expiration date set in the past,

```
<?php
// Deleting a cookie
setcookie("username", "", time()-3600);
?>
```

# Scope of cookies

- Cookies can only be read from the site from which they were set
- this helps to ensure that one can not steal cookies (and thus identities) through hostile websites

# Cookies and their use

- Shopping cart
  - when the front page appears, set a new (empty) cookie
  - items are added by updating the cookie
  - *alternatively, one can store goods in the server's database and just store an ID in the cookie that points to your basket*
- Login
  - user inputs name and password into a form
  - *after the combination has been verified, it sends a cookie to the user that identifies the user to the system*
  - next time the user visits the page, the web server checks if there is a cookie, and if so the user is identified

# Criticism of cookies

- One can not be completely anonymous on the net
  - most ads/banners come from relatively few advertisers.
  - these may, with the help of cookies, follow a browser/ computer combination on all the sites they advertise on.
- there have been examples of security vulnerabilities in browsers, so that adversaries can get access to cookies
  - which they can use to gain access to sites with a faked identity

# Sessions

- A combination of cookies and data stored on the server (automatically by PHP)
  - saves a cookie containing an ID on the user's computer that points to a session on the server
- A session is a global PHP array ($_SESSION)
- A session is designed as an easy way to store data – for a short period
  - a session's lifetime in PHP is only 24 minutes by default

# Starting PHP session

- Sessions must be started at the top of the page before it is used

  **session_start()**

- Session_start() function creates a new session and generate a unique session ID for the user.
  - it first checks for an existing session ID. If it finds one, i.e. if the session is already started, it sets up the session variables and if doesn't, it starts a new session by creating a new session ID.

# Storing and Accessing Session Data

- session data can be stored as key-value pairs in the $_SESSION[] super global array.
  - The stored data can be accessed during lifetime of a session.

```
<?php
// Starting session
session_start();

// Storing session data
$_SESSION["firstname"] = “Abebe";
$_SESSION["lastname"] = “Lemlem";
?>
```

# Accessing session data

- We can access the session data we set on our previous example from any other page on the same web domain
- Simply recreate the session by calling session_start() and then pass the corresponding key to the $_SESSION associative array.

```
<?php
session_start();
echo $_SESSION["Name"];
?>
```

# Destroying PHP session

- all global session variables can be removed by destroing the session using
  - session_unset(“sessionId”) //remove all session variables
  - session_destroy():// destroy the session

```
<?php
session_start();
// Removing session data
if(isset($_SESSION[“username"])){
session_unset($_SESSION[“usernam
e"]);
}
?>
```

```
<?php
session_start();
// Destroying session
session_destroy();
?>
```

# Session for login

- Sessions are tied to an individual user and a corresponding browser
- therefore, they are well suited to handle logins
- Useful command in this context: header()
- **header()** *is used to send HTTP headers to the browser*
- *as other header information (such as cookies and sessions), such a command should precede any HTML in a PHP file*
- *a particularly interesting header in this context is* **Location:**, *which redirects the browser to another URL*

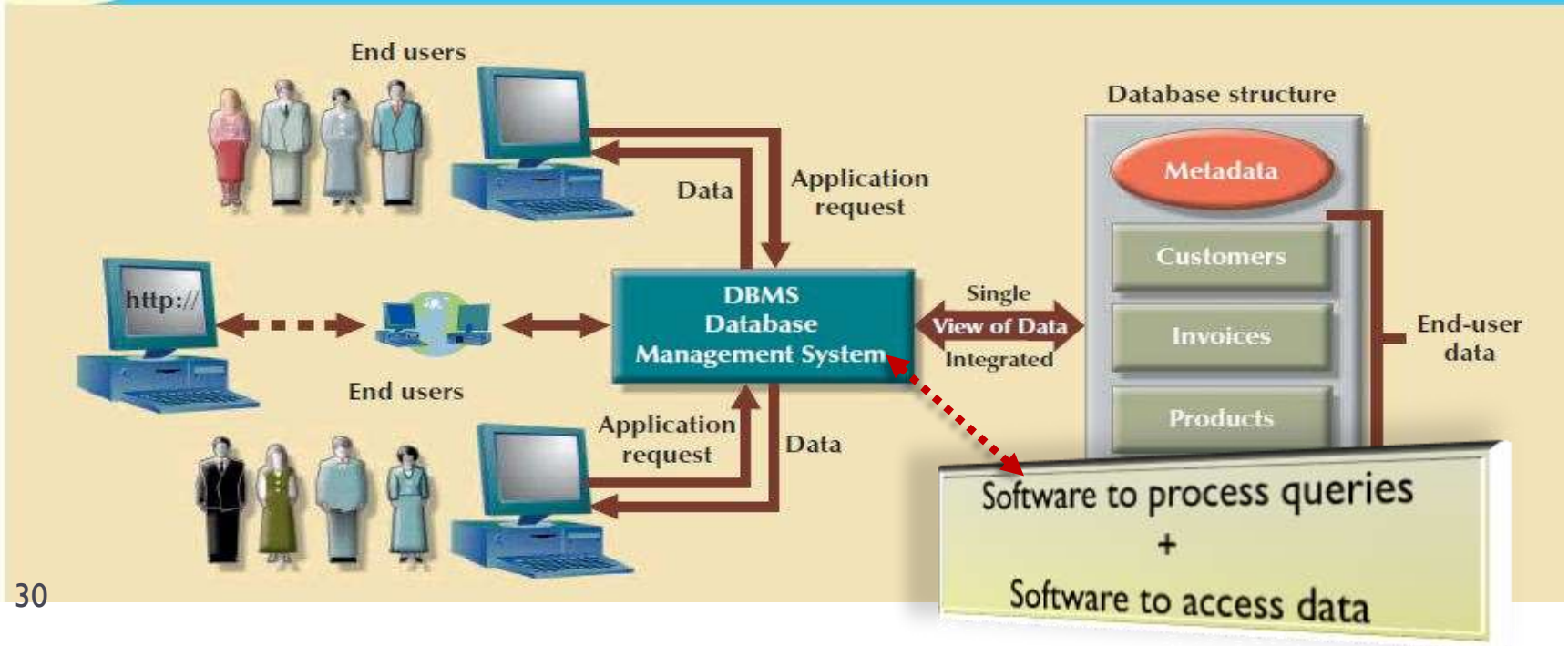
```php
<?php
session_start();
echo session_id() . '<br>';
if(isset($_SESSION['counter'])){
    $_SESSION['counter']++;
}else{
    $_SESSION['counter']=1;
    }
?>

<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
          content="width=device-width, user-scalable=no, initial-scale=1.0, maximum-scale=1.0, minimum-scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <title>Document</title>
</head>
<body>
<h1>Page visited:<?php echo $_SESSION['counter'] ?> Times</h1>
</body>
</html>
```

# Data Base manipulation using PHP

# Review on Database terms

- **Database:** is a collection of related data
- **Database Management System (DBMS)**: A software package/ system to facilitate the creation and maintenance of a computerized database.

FIGURE The DBMS manages the interaction between the end user and the database



# Databases and PHP

- Currently PHP can communicate with different DBMS's

Connection with database can be achieved either through

  - Vendor specific extensions such as MySQL Drivers and Plugins or
  - DAB's -database abstraction layers such as, PDO, ODBC, dbx, DBA

# Introduction to MySQL

- MySQL is a RDBMS
- Is an open-source software relational database management system
- Use standard form of SQL data language

- Assignemt – 1 : submission deadline Jan 20, 2021
  - Select one NoSQL database of your preference and prepare a hand written note with not more than 2 pages on the features of the DB and compare it to that of relational database
  - Consider the selected NoSQL database and write a working PHP code to INSERT, UPDATE, DELETE and SELECT(retrive a data)

# Basic queries

- CREATE: create databases and tables
  - *CREATE TABLE tableName(columnName Datatype constraint, …. );*
- SELECT: select table rows based on certain conditions
  - *SELECT columnName, columnName… FROM tableName WHRER condition;*
- DELETE: delete one or more rows of a table
  - *DELETE FROM tableName WHERE condition;*
- INSERT: insert a new row in a table
  - *INSERT INTO tableName(column list) VALUES (column values);*
- UPDATE: update rows in a table
  - *UPDATE tableName SET columnName = NewValues;*
- ALTER: alter the structure of a table
  - *ALTER TABLE tableName ADD columnBame dataType;*
  - *ALTER TABLE tableName DROP columnName;*
  - *ALTER TABLE tableName MODIFY columnName Datatype [NULL value] [DEFAULT value];*

# Administrative MySQL Command:

- **USE *Databasename***: This will be used to select a particular database in MySQL work area.
- **SHOW DATABASES:** Lists the databases that are accessible by the MySQL DBMS.
- **SHOW TABLES:** Shows the tables in the database once a database has been selected with the use command.
- **SHOW COLUMNS FROM *tablename:*** Shows the attributes, types of attributes, key information, whether NULL is permitted, defaults, and other information for a table.
- **SHOW INDEX FROM *tablename*:** Presents the details of all indexes on the table, including the PRIMARY KEY.
- **SHOW TABLE STATUS LIKE *tablename*\G:** Reports details of the MySQL DBMS performance and statistics.
- **Create user:** CREATE USER 'newuser'@'localhost' IDENTIFIED BY 'password';

**Grant privilege:** GRANT type_of_permission ON database_name.table_name TO 'username'@'localhost';

**Reload privileges:** FLUSH PRIVILEGES;

**revoke permission:** REVOKE type_of_permission ON database_name.table_name FROM 'username'@'localhost';

**View permission:** SHOW GRANTS FOR 'username'@'localhost';

# PHP and MySQL common functions

- Connecting to a Database
- Making a query
- Using results of a query
- Freeing resources
- closing the connection

# Connecting to a Database

- Opens a new connection MySQL server
- Can be done through mysqli_connect(procedural style), ***mysqli*** (OOP) or by data abstraction layer (***PDO***)

```php
<?php
function connect_db($dbname)
{
define("HOST", "localhost");
define("USER", "root");
define("PWD", "passwd");
define("DB_NAME", $dbname);

$mysqli = new mysqli(HOST, USER, PWD, DB_NAME);

if($mysqli->connect_errno){
    echo "connection failed: " . $mysqli->connect_error;
    exit();
}
return $mysqli;
}
?>
```

```php
public PDO $pdo;

public function __construct()
{
    $this->pdo = new PDO('mysql:server=localhost;dbname=notes','root', 'passwd');
    $this->pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
}
```

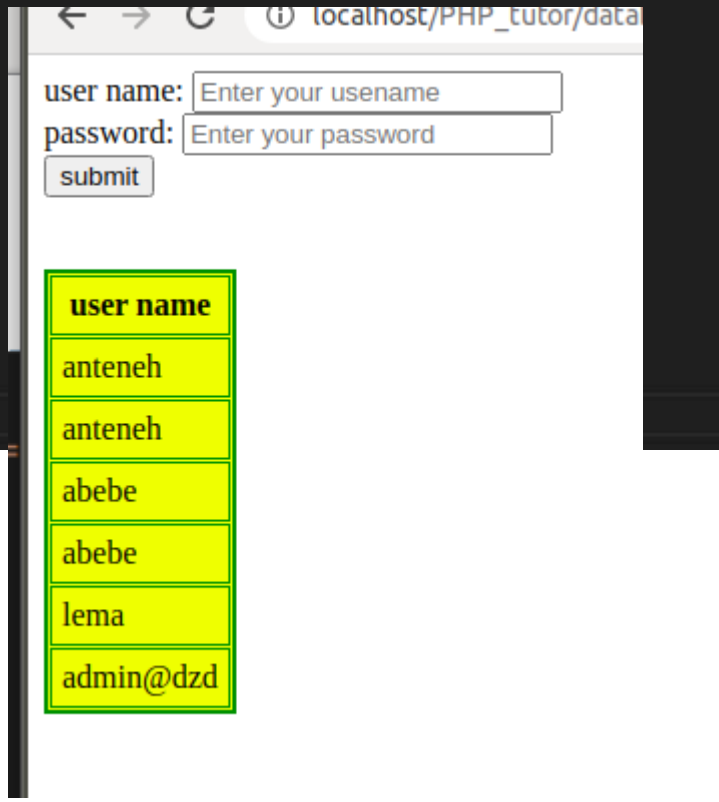
# Making a query (*1. SELECT*)

```
function view(){
$con = connect_db('notes');
$query = "SELECT * FROM users";
$result = $con->query($query);
return $result;
```

```
public function getNotes(){
    $statment = $this->pdo->prepare("SELECT * FROM notes ORDER BY create_date");
    $statment->execute();
    return $statment->fetchAll(PDO::FETCH_ASSOC);
}
```

# Using the query result : example

```php
<?php
echo '<table border="2" cellpadding="5" cellspacing="1" bordercolor ="green" bgcolor="yellow">';
echo '<tr><th> user name</th></tr>';

include('select_mysqli.php');

$result = view();

while($r= $result->fetch_assoc())
{
    echo '<tr><td>'. $r['username'] . '</td></tr>';
}
echo '</table>';
?>
```



# Making a query : insert

```
$con = connect_db('notes');
$uname=$con->real_escape_string($_POST['username']);
$passwd = $con->real_escape_string($_POST['password']);

$query = "INSERT INTO users(username, password) VALUES('$uname', '$passwd')";
$result = $con->query($query);
if(!$result)
{
    $err =1;
}

$con->close();

if(isset($err))
{
    header("location: index.php?err=1");
}
else
{
    header("location: index.php");
}
```

# Making a query : insert, update - PDO

```php
public function addNote($note) {

    $statment = $this->pdo->prepare("INSERT INTO notes(title, description, create_date) VALUES(:title, :description, :date)");
    $statment->bindValue('title', $note['title']);
    $statment->bindValue('description', $note['description']);
    $statment->bindValue('date', date('Y-m-d H:i:s'));
    return $statment->execute();
}
```

```php
public function updateNote($id, $note){

    $statment= $this->pdo->prepare("UPDATE notes SET title = :title, description = :description WHERE id= :id");
    $statment->bindValue('id', $id);
    $statment->bindValue('title', $note['title']);
    $statment->bindValue('description', $note['description']);
    return $statment->execute();
}

public function deleteNote($id){
    $statment= $this->pdo->prepare("DELETE FROM notes WHERE id= :id");
    $statment->bindValue('id', $id);
    return $statment->execute();


}
```

# Freeing query resources and closing connection

- free memory associated with the given resource
- called result (after a select query).
- Not necessary except for large result sets
- Done automatically when script exits.

closing

```
$result->close();
```

: close the database connection associated with the given database link

–

```
$con->close($con)
```

# Other functions

**$con->real_escape_string($toEscapeString);**

- This function is used to create a legal SQL string that you can use in an SQL statement.
- The given string is encoded to an escaped SQL string, taking into account the current character set of the connection.
- parameters
  - $toEscapeString: the string to be escaped.
    - Characters encoded are NUL (ASCII 0), \n, \r, \, ', ", and Control-Z.

# Error handling

## mysqli_connect_errno()

- Returns the last error code number from the last call to mysqli_connect
- An error code value for the last call to mysqli_connect, if it failed.
- zero means no error occurred

## mysqli_connect_error

- Returns a string description of the last connect error
- NULL is returned if no error occurred.

# Reference

- For more detail review the PHP manual https://www.php.net/manual/en/refs.database.php
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